of resource-rich countries with SWFs: over time, wealth is transferred from human capital to financial wealth, just as Timor-Leste transfers wealth from under the sea to its Petroleum Fund. When human capital is exhausted (the oil is gone), we consume from retirement savings (or the SWF). This *life-cycle* profile affects how we need to save; the composition of our financial portfolios changes as we age (see chapter 5). The fact that we can lose our labor income over our working years, hopefully just temporarily, leads us to hold portfolios that cushion the fall when we lose our jobs.

2. Leverage

   Borrowing enables us to smooth consumption, but leverage leads to increased risk. Housing, the largest asset position for individuals, is typically highly levered and also highly illiquid. Leverage is a short position in bonds, and so the middle class is highly exposed to *interest rate risk* (or *duration risk*). Illiquidity risk also carries its own consideration (see chapter 13).

3. Health care

   A bad health shock can be disastrous. The government, through Medicare and Medicaid, provides some support for health care, but it is highly incomplete. We can buy health and disability insurance to hedge some of this risk, but we cannot completely remove the effects of these *idiosyncratic shocks*.[74] The presence of such *background risk* leads us to be effectively more risk averse than would be the case if we were guaranteed perfect health at all times.

And what about the poor? Studies show that financial education actually makes little difference for the poor: what's the point of good financial knowledge if you don't have any money to invest in the first place?[75] Addressing the (in)adequacy of savings for our poor and even middle class involve national-level savings systems to deal with bad bouts of unemployment, health-care shocks, and for retirement. There have been many proposals to reform our national pension system and many involve an increased role for government.[76] This takes us back where we started—to the management of SWFs.

6. Timor-Leste Redux

According to the IMF, Timor-Leste is "the most oil-dependent economy in the world."[77] The government obtains 95% of its revenue from oil and gas, and the

[74] This is the notion of Kimball (1990), where any unexpected, uninsurable risk raises the demand for safe assets.

[75] Lusardi, Michaud, and Mitchell (2013) show that if you are poor enough, it is optimal to remain financially illiterate over your whole life.

[76] One that is surprisingly contentious is by Teresa Ghilarducci (2008), who recommends introducing compulsory savings levies managed by Social Security.

[77] IMF Executive Board Concludes 2010 Article IV Consultation with the Democratic Republic of Timor-Leste, IMF Public Information Notice No. 11/31, March 8, 2011.